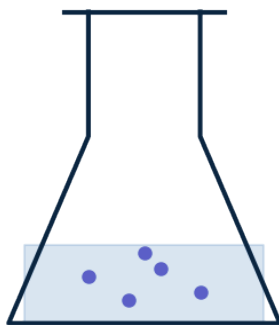


Mole Concept

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- A mole is the amount of substance containing as many elementary entities as there are atoms in 12 g of carbon-12: $N_A = 6.022 \times 10^{23}$ per mole (Avogadro's number).
- Molar mass is the mass of one mole of a substance in grams, numerically equal to the atomic/molecular/formula mass in amu.
- Number of moles $n = \text{given mass} / \text{molar mass} = \text{number of particles} / N_A = \text{volume of gas at STP} / 22.4 \text{ L}$.
- Percentage composition of an element in a compound = $(\text{mass of element in one mole of compound} / \text{molar mass of compound}) \times 100$.
- Empirical formula shows the simplest whole-number ratio of atoms; molecular formula = $(\text{empirical formula})_n$, where n is a whole number found from molar mass.
- Limiting reagent is the reactant that is completely consumed first, limiting the amount of product formed; other reactants are said to be in excess.
- Molarity (M) = moles of solute / volume of solution in liters. Molality (m) = moles of solute / mass of solvent in kg.



1 mole = 6.022×10^{23} particles (Avogadro's number)

One mole of any substance contains 6.022×10^{23} particles (Avogadro's number).

2. Formula Sheet

Number of moles	$n = \text{mass} / \text{molar mass}$
Moles from particles	$n = N / N_A$
Moles of gas at STP	$n = V / 22.4 \text{ L}$
Molarity	$M = \text{moles of solute} / \text{volume (L)}$
Molality	$m = \text{moles of solute} / \text{mass of solvent (kg)}$
Mole fraction	$x_A = n_A / (n_A + n_B)$
% composition of element	$(\text{mass of element in 1 mol} / \text{molar mass}) \times 100$
Dilution law	$M_1 V_1 = M_2 V_2$

3. Practice Problems (22 questions)

- Find the number of moles in 44 g of CO_2 (molar mass 44 g/mol).
- Find the number of molecules in 2 moles of water ($N_A = 6.022 \times 10^{23}$).
- Find the volume occupied by 0.5 mole of an ideal gas at STP.
- Calculate the molarity of a solution containing 40 g of NaOH (molar mass 40 g/mol) in 2 L of solution.
- An oxide of nitrogen has the empirical formula NO_2 and a molar mass of 92 g/mol. Find its molecular formula.
- Find the percentage of oxygen by mass in water (H_2O , molar mass 18 g/mol).
- 10 g of CaCO_3 (molar mass 100 g/mol) is heated to decompose completely into CaO and CO_2 . Find the mass of CO_2 produced.
- In the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, if 2 mol of N_2 reacts with 3 mol of H_2 , identify the limiting reagent.
- Find the mass of 3 moles of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$, molar mass 180 g/mol).
- How many grams of NaCl (molar mass 58.5 g/mol) are needed to prepare 500 mL of a 0.5 M solution?
- Find the mole fraction of solute in a solution containing 2 mol of solute and 8 mol of solvent.
- State the difference between empirical formula and molecular formula, with one example.
- A compound contains 40% carbon, 6.7% hydrogen, and 53.3% oxygen by mass. Find its empirical formula (atomic masses: $\text{C}=12$, $\text{H}=1$, $\text{O}=16$).
- Find the number of moles of oxygen atoms in 3 moles of CO_2 .
- 500 mL of a 2 M solution is diluted to 2 L. Find the new molarity.
- Explain why the mole concept is essential in stoichiometric calculations of chemical reactions.
- Find the mass in grams of a single atom of carbon-12 ($N_A = 6.022 \times 10^{23}/\text{mol}$).

18. In the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, find the number of moles of water formed from 4 moles of H_2 (assuming excess O_2).
19. Find the number of moles present in 5.6 L of a gas at STP.
20. State Avogadro's law and explain how it relates the volume of a gas to the number of moles at fixed T and P.
21. A 250 mL solution contains 0.1 mol of KCl. Find its molarity.
22. Explain what is meant by a 'limiting reagent' and how it determines the theoretical yield of a reaction.

Answer Key

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| 1. $n = 1 \text{ mol}$ | 2. $N = 1.2044 \times 10^{24} \text{ molecules}$ |
| 3. $V = 11.2 \text{ L}$ | 4. $M = 0.5 \text{ mol/L (0.5 M)}$ |
| 5. Molecular formula = N_2O_4 | 6. % O = 88.9% |
| 7. Mass of $\text{CO}_2 = 4.4 \text{ g}$ | 8. N_2 is the limiting reagent (needs $3 \times 3 = 9 \text{ mol H}_2$ but only 3 mol available relative to 2 mol $\text{N}_2 \rightarrow \text{H}_2$ is actually limiting; re-check: 2 mol N_2 needs 6 mol H_2 , only 3 mol H_2 given, so H_2 is limiting) |
| 9. Mass = 540 g | 10. Mass $\text{NaCl} = 14.6 \text{ g}$ |
| 11. Mole fraction = 0.2 | 12. Conceptual - empirical formula gives simplest ratio (e.g., CH_2O for glucose); molecular formula gives actual composition ($\text{C}_6\text{H}_{12}\text{O}_6$) |
| 13. Empirical formula = CH_2O | 14. Moles of O atoms = 6 mol |
| 15. New molarity = 0.5 M | 16. Conceptual - moles link mass, particles, and volume, enabling balanced-equation calculations |
| 17. Mass = $1.99 \times 10^{-23} \text{ g}$ | 18. Moles of water = 4 mol |
| 19. $n = 0.25 \text{ mol}$ | 20. Conceptual - equal volumes of gases at the same T and P contain equal numbers of moles |
| 21. $M = 0.4 \text{ mol/L}$ | 22. Conceptual - limiting reagent is used up first and caps the maximum product possible |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.